Parsa Esmkhani

709-986-6515 | pesmkhani@mun.ca | parsaesmkhani.github.io | linkedin.com/in/parsaesmkhani

Education

B.Sc. (Honours) in Computer Science — Sep 2021 – Apr 2027 (Exp.)
Minors in Mathematics and Psychology — *Memorial University, St. John's, NL*

- **GPA:** 3.97/4.00 — **Concentration:** Theory of Computation
- **Honours Thesis:** *Game-Theoretic De-randomisation and Pseudorandom Generators*
- **Relevant Courses (Grade):** COMP 4742 Computational Complexity (100), ECON 4200 Game Theory (100), COMP 3602 Theory of Computation (97), PHIL 2030 Logic (96), PHIL 2031 Intermediate Logic (95), COMP 1002 Logic for CS (100), COMP 3600 Algorithm Design & Analysis (91), COMP 3200 Algorithmic Techniques for AI (92)

Research Experience

Research Assistantships — Sep 2025 – Present
Departments of Neuroscience, Geography, Economics, Business, and Linguistics — *Memorial University*

- **Computational Neuroscience** (Sep 2026–Present; Dr. Qi Yuan): Developing computational strategies for multi-omics analysis of an Alzheimer's disease rat model, integrating microbiome, metabolomic, lipidomic, brain transcriptomic, and behavioral data to investigate gut–brain relationships
- **Computational Sociolinguistics** (Sep 2026–Present; Dr. Paul De Decker): Developing a Godot-based 3D fieldwork simulator for undergraduate linguistics education to explore Newfoundland English variation and research ethics
- **Urban Ecology & Health** (Sep 2026–Present; Dr. Mahyar Masoudi): Analyzing national Statistics Canada data on excess heat and well-being across Canadian cities, including data cleaning, management, and analysis in Python.
- **Mathematical Economics** (May 2026–Present; Dr. Jean-Gabriel Lauzier): Adapting deep neural network methods for optimal allocation among Friedman-Savage agents in non-convex risk-sharing problems
- **Geography** (Jun 2026–Aug 2026; Dr. Nicholas Lynch): Collected and synthesized literature and web-scraped data to support research on demolition-material reuse, expanding the evidence base for circular-economy analyses
- **Business** (Sep 2025–July 2026; Dr. Chansoo Park): Building and validating a large-scale corporate divestment and exit dataset from Factiva, supporting analysis of news coverage, sentiment, and firm exit decisions

Summer Undergraduate Research Award (SURA) — May 2026 – Aug 2026
Faculty of Medicine, Dr. Michael Wahl — *Memorial University*

- Developed a Python transcription workflow and SQLite database for 78 qualitative interviews, creating a searchable corpus for downstream machine-learning-based information extraction

Honours Research Thesis | *Supervisor: Dr. Antonina Kolokolova* — Fall 2025 – Winter 2026

- Proved a derandomization result showing that, under suitable assumptions, the existence of a pseudorandom generator allows Reality to keep Skeptic's capital bounded in a variant of the Skeptic–Reality game

Awards

- **1st Place, IBM watsonx Hackathon (2026)** — Winning team among 16 teams for *B'y Your Side*
- **Summer Undergraduate Research Award (SURA) (2026)** — Received $3,000 in research funding under the supervision of Dr. Michael Wahl. (Internal analogue of the NSERC USRA Award)
- **Dean's List (2023–2024, 2024–2025)** — Top 10% of Faculty of Science students based on GPA

Conferences & Summer Schools

- **Carnegie Mellon University Summer School in Logic and Formal Epistemology (2026)** — One of 25 students selected globally; unable to attend due to U.S. visa restrictions affecting Iranian nationals
- **York University Centre for Vision Research (CVR) Vision Science Summer School (2026)** — One of 20 students selected globally; travel and accommodation funded by York University
- **ISM Discovery School: Complexity and Mathematical Logic (2026)** — Received $1,500 in travel funding to attend the program at *Concordia University* in Montreal
- **Computational Complexity Conference (CCC) (2025)** — Received $2,100 in travel funding ($300 from CCC and $1,800 from supervisor) to attend the conference at the *Fields Institute, University of Toronto*

Presentations

- **Familiarity in Humans vs. Artificial Neural Networks** — CVR Summer School, York University (Jun 12, 2026)
- **Derandomization in a Skeptic–Reality Game** — SEA Conference, Memorial University (Apr 9, 2026)
- **Fairness in Resource Allocation; Cake Cutting** — Theory Group Seminar, Memorial University (Mar 11/18, 2025)

Teaching Experience

Peer Assisted Learning [PAL] Leader for COMP 1002 (Logic) Winter 2023 – Present
Academic Success Center *Memorial University*
- Conducted two weekly PAL tutorials for 12 semesters reviewing course context and preparing for exams
- Accumulated the highest number of student contact hours among all PAL subjects, totaling 1,700+ hours

Teaching Assistant — COMP 1002: Introduction to Logic Winter 2026
Department of Computer Science *Memorial University*
- Provided instructional support for ∼60 students during weekly labs, as well as grading and exam invigilation

Computer Science Help Centre Assistant Fall 2025/Winter 2026
Department of Computer Science *Memorial University*
- Provided support for Computer Science students in year 1 and 2 for COMP 1001, 1002, 1003, 2001, 2002, and 2003

Substitute Instructor — COMP 1002: Introduction to Logic Fall 2025/Winter 2026
Department of Computer Science *Memorial University*
- Delivered four lectures in place of the course instructor on induction, structural induction, FOL logic, and relations

Residence Computer Science/Mathematics Assistant Fall 2025
Hatcher Commons – Student Residences *Memorial University*
- Provided on-site assistance for residence students in first-year Computer Science and Mathematics courses

Other Work Experience

Persian/Dari Interpreter | *Association for New Canadians; Kelly Services Canada* Oct 2021 – Jan 2026
- 6000+ hours of medical, banking, and educational interpretation; trained by Language Line Solutions.

WUSC Ambassador | *Internationalization Office, Memorial University* Fall 2021 – Spring 2022
- Supported a World University Service of Canada scholar's settlement and organized events for MUN WUSC chapter.

Projects

IntuitiveCS.ca | *Python, ReactFlow, CSS, HTML, JavaScript* Dec 2025 – Present
- Co-developed a website featuring interactive visualizations of mathematical computer science concepts for logic

B'y Your Side | *IBM watsonx, Healthcare AI, LLMs, Wearables* May 29–31, 2026
Co-developed a healthcare AI assistant integrating patient records and wearable data for health alerts (Video Demo)

COMP 1002 Interactive Logic Exercises | *Carnap, Markdown* Spring 2022
- Assisted Dr. McIntyre in developing proof-checking tools and interactive exercises (Repository)

Academic Leadership & Service

- **Undergraduate Committee Student Rep** | *Department of Computer Science* Sep 2025 – Present
- **Co-Organiser of Theory Group** | *Department of Computer Science* Sep 2023 – Present
- **Student Rep, Department Head Review Committee** | *Department of Computer Science* May 2026 – Aug 2026
- **Science Student Rep, Annual Student Recruitment Panel** | *Faculty of Science* Feb 2026
- **Faculty of Science Student Rep for Computer Science** | *Faculty of Science* Jan 2024 – May 2025
- **Computer Science Society Executive (Social Outreach)** | *Memorial University* Jan 2022 – Dec 2025

Volunteering

- **Everything Science Volunteer** | *Memorial University, St. John's, NL* Jan 2024 – Present
- **SEA Conference Volunteer** | *Memorial University, St. John's, NL* Apr 2024, 2025, 2026
- **Canada Games 2025 Volunteer (Venue Technology Crew)** | *Canada Games, St. John's, NL* Aug 2025